

DHR/DR Sector Reconstruction Does Not Invert a Nonfaithful Restriction

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Abstract

Doplicher–Haag–Roberts and Doplicher–Roberts reconstruction theorems start from a supplied superselection sector structure. They reconstruct the field algebra and compact gauge group associated with that supplied structure. This note isolates the finite-central boundary: if a finite central amplification

$$\mathcal{A}^{(n)}(O) = \mathcal{A}(O) \bar{\otimes} \ell^\infty(\{1, \dots, n\})$$

is first restricted to the diagonal subnet $a \mapsto a \otimes 1$, then the central-character sectors over a fixed sector become identical in the supplied restricted category. Applying DHR/DR reconstruction after that restriction cannot recover the central characters, the central cardinality n , or the predicate $n = 2$. The sharp positive exit is sector-faithfulness: the sector datum supplied to reconstruction must be injective on the finite central-character fibers, or the theory must explicitly quotient, factor, or exclude those fibers. If the full character-labeled sector category is retained instead, the finite central cardinality is already readable as the cardinality of the retained character fiber.

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1 Setup

Let $O \mapsto \mathcal{A}(O)$ be a Haag–Kastler net and let $\mathcal{S}(\mathcal{A})$ be a chosen DHR-type sector category of representations of $\mathcal{A}$. For $n \geq 2$, define the finite central amplification

$$\mathcal{A}^{(n)}(O) = \mathcal{A}(O) \bar{\otimes} \ell^\infty(\{1, \dots, n\}).$$

Let e_i be the minimal central projections of $\ell^\infty(\{1, \dots, n\})$, and let $\chi_i : \ell^\infty(\{1, \dots, n\}) \rightarrow \mathbb{C}$ be the corresponding characters. The diagonal subnet is the inclusion

$$\iota_O : \mathcal{A}(O) \rightarrow \mathcal{A}^{(n)}(O), \quad \iota_O(a) = a \otimes 1.$$

For an object $\pi \in \mathcal{S}(\mathcal{A})$, define a representation π_i of $\mathcal{A}^{(n)}$ by

$$\pi_{i,O}(a \otimes z) = \pi_O(a)\chi_i(z).$$

Let $\mathcal{S}_{\text{char}}^{(n)}$ be the full sector category generated by these objects π_i .

Theorem 1 (sector-character restriction fiber). *The category $\mathcal{S}_{\text{char}}^{(n)}$ is the disjoint union of n copies of $\mathcal{S}(\mathcal{A})$. More precisely,*

$$\text{Hom}_{\mathcal{S}_{\text{char}}^{(n)}}(\pi_i, \rho_j) = \begin{cases} \text{Hom}_{\mathcal{S}(\mathcal{A})}(\pi, \rho), & i = j, \\ 0, & i \neq j. \end{cases}$$

Restriction to the diagonal subnet sends every π_i to π . Thus the restricted sector datum is the projection

$$\mathcal{S}(\mathcal{A}) \times \{1, \dots, n\} \longrightarrow \mathcal{S}(\mathcal{A}), \quad (\pi, i) \longmapsto \pi,$$

and it is not faithful on finite central-character fibers.

Proof. For $i = j$, an intertwiner $T : \pi_i \rightarrow \rho_i$ satisfies

$$T\pi_O(a)\chi_i(z) = \rho_O(a)\chi_i(z)T$$

for all $a \otimes z$. Taking $z = 1$ shows that T is exactly an intertwiner $\pi \rightarrow \rho$ in $\mathcal{S}(\mathcal{A})$, and every such intertwiner plainly intertwines π_i and ρ_i .

For $i \neq j$, if $T : \pi_i \rightarrow \rho_j$, then

$$T = \pi_i(1 \otimes e_i)T = T\rho_j(1 \otimes e_i) = 0,$$

since $\pi_i(1 \otimes e_i) = 1$ while $\rho_j(1 \otimes e_i) = 0$. Hence different central characters have no intertwiners. Finally,

$$\pi_i(\iota_O(a)) = \pi_i(a \otimes 1) = \pi_O(a),$$

so diagonal restriction forgets i . □

Theorem 2 (sharp categorical exit). *Let $\mathcal{C}$ be a sector category before a possible restriction, let*

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

be the sector datum actually supplied to a reconstruction procedure, and let $\kappa : \text{Obj}(\mathcal{C})/\simeq \rightarrow X$ be a finite central-character label. Then κ is reconstructible from F if and only if

$$F(\rho) \simeq F(\sigma) \implies \kappa(\rho) = \kappa(\sigma) \tag{1}$$

for all objects $\rho, \sigma \in \mathcal{C}$. Equivalently, the supplied sector datum must be faithful on the fibers that κ separates.

Proof. If κ is reconstructible from F , then $\kappa = \widehat{\kappa} \circ F$ on isomorphism classes, so (1) follows. If (1) holds, define $\widehat{\kappa}$ on the image of F by choosing any ρ with prescribed image and setting $\widehat{\kappa}(F(\rho)) = \kappa(\rho)$. Condition (1) is exactly the well-definedness of this assignment. □

Proposition 1 (full character fiber recovers cardinality). *Assume the supplied sector datum retains the full central-character category $\mathcal{S}(\mathcal{A}) \times F$, rather than its diagonal projection, and that the projection to the base sector category is part of the supplied comparison data. For any base object π , the retained fiber*

$$\{\rho \in \text{Obj}(\mathcal{S}(\mathcal{A}) \times F) / \simeq : \rho \mapsto \pi\}$$

has cardinality $|F|$. Hence the finite central cardinality is reconstructible from the supplied category-with-character-fiber data, and the predicate $|F| = 2$ is reconstructible exactly when this retained fiber has two objects.

Proof. In the full category, the objects over π are precisely (π, i) for $i \in F$. Theorem 1 shows that objects with different characters have no intertwiners, hence are nonisomorphic, while same-character copies have the usual intertwiners of $\mathcal{S}(\mathcal{A})$. Therefore the retained fiber has exactly one isomorphism class for each $i \in F$, so its cardinality is $|F|$. If the projection to the base sector category is retained, this fiber is a function of the supplied data. If the category is instead replaced by the diagonal restricted category, the fiber is collapsed before reconstruction and its cardinality is no longer a function of the supplied datum. $\square$

2 DHR/DR Boundary

DHR analysis supplies localized superselection sectors under its selection criterion. Doplicher–Roberts reconstruction and compact-group duality then apply to the symmetric tensor C^* -category and fiber structure that are actually supplied.

Proposition 2 (DHR/DR does not recreate omitted central characters). *Suppose the sector data supplied to DHR/DR reconstruction are formed only after restricting the finite central amplification $\mathcal{A}^{(n)}$ to the diagonal subnet $\mathcal{A} \otimes 1$. Then, for every $n \geq 2$, the supplied restricted sector category is equivalent to $\mathcal{S}(\mathcal{A})$. Consequently DHR/DR reconstruction from that restricted category cannot determine the central-character label i , the finite central cardinality n , or the two-atom predicate $n = 2$.*

Proof. By Theorem 1, the full central-character sector category is $\mathcal{S}(\mathcal{A}) \times \{1, \dots, n\}$, and diagonal restriction is the projection to $\mathcal{S}(\mathcal{A})$. Therefore the category supplied after restriction is equivalent to $\mathcal{S}(\mathcal{A})$, independently of n . Invariants of the supplied restricted category, including any reconstruction procedure whose input is that category, cannot distinguish $n = 2$ from $n = 3$, nor can they distinguish the central characters over a fixed π . The sharp exit is Theorem 2: retain a sector datum faithful on these fibers, or exclude/quotient the amplified comparison class. $\square$

3 Positive Exits

Exit premise	Effect
Full sector category retained	If the supplied category is $\mathcal{S}(\mathcal{A}) \times \{1, \dots, n\}$, the finite central characters are part of the input, the retained character fiber has cardinality n by Proposition 1, and DHR/DR is not being asked to recover forgotten labels.
Faithful sector functor	A restriction or forgetful datum F is sufficient exactly when it is injective on the central-character fibers relevant to the target label.
Central-character readout	Minimal central projections, central characters, or equivalent sector labels separate the i -fibers.
Quotient or exclusion	If the finite central amplification is explicitly quotiented, factor-valuedness is imposed, or such sectors are physically inadmissible, then the comparison class is removed.

4 Referee Question

The classification question is precise:

Does the DHR/DR sector structure being supplied in the target setting already retain finite central-character fibers, or impose a quotient/factor/exclusion axiom that removes them? If not, reconstruction from the supplied restricted category does not certify finite-center accountability or binary rank.

The desired external response is one of: known theorem, false statement, missing positive hypothesis, physically irrelevant comparison class, or useful expository no-go.

References

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